10th grade
Learning evidence 2.1: Analysis of Decisions (Solutions)

Problems

Problem 1: Structured solution

Interpreting decision alternatives, events, consequences, and states of nature

Element	Description
Decision alternatives	Plan A (local produce), Plan B (mixed menu), Plan C (bulk staples)
States of nature	High demand (0.35), Medium demand (0.45), Low demand (0.20)
Events	Demand level observed after selecting a plan
Consequences	Profit (hundreds of dollars) for each plan–demand pair
Probabilities	0.35, 0.45, 0.20 for high, medium, low demand

Building the payoff table from the description

State of nature	Probability	Plan A	Plan B	Plan C
High demand	0.35	92	70	55
Medium demand	0.45	50	62	48
Low demand	0.20	-8	30	38

Applying the Maximax criterion We ignore probabilities and select the plan with the highest possible payoff.

$$\max_i \text{Payoff}(\text{Plan A}, S_i) = 92$$
$$\max_i \text{Payoff}(\text{Plan B}, S_i) = 70$$
$$\max_i \text{Payoff}(\text{Plan C}, S_i) = 55$$
$$\max\{92, 70, 55\} = 92$$

The Maximax rule recommends Plan A.

Applying the Maximin criterion We focus on the worst payoff for each plan.

$$\min_i \text{Payoff}(\text{Plan A}, S_i) = -8$$
$$\min_i \text{Payoff}(\text{Plan B}, S_i) = 30$$
$$\min_i \text{Payoff}(\text{Plan C}, S_i) = 38$$
$$\max\{-8, 30, 38\} = 38$$

The Maximin rule recommends Plan C.

Formulating the decision problem for an optimal choice We compute expected profits using the stated probabilities.

$$\begin{aligned} EV_A &= 0{,}35(92) + 0{,}45(50) + 0{,}20(-8) \\ &= 32{,}2 + 22{,}5 - 1{,}6 = 53{,}1 \\ EV_B &= 0{,}35(70) + 0{,}45(62) + 0{,}20(30) \\ &= 24{,}5 + 27{,}9 + 6 = 58{,}4 \\ EV_C &= 0{,}35(55) + 0{,}45(48) + 0{,}20(38) \\ &= 19{,}25 + 21{,}6 + 7{,}6 = 48{,}45 \end{aligned}$$

$$\text{máx}\{53{,}1, 58{,}4, 48{,}45\} = 58{,}4$$

The expected value criterion recommends Plan B.

Problem 2: Structured solution

Interpreting decision alternatives, events, consequences, and states of nature

Element	Description
Decision alternatives	Strategy A (electric), Strategy B (hybrid), Strategy C (diesel)
States of nature	Combined demand and fuel-cost outcomes (6 states)
Events	Demand level and fuel cost realized after the strategy choice
Consequences	Profit (thousands of dollars) for each strategy–state pair
Probabilities	Demand: 0.30, 0.50, 0.20; Fuel cost: 0.55, 0.45; independent

Building the payoff table from revenue and cost data Combined state probabilities are obtained by multiplying demand and fuel-cost probabilities.

State of nature	Probability	Strategy A	Strategy B	Strategy C
Strong demand, low cost	0.165	260	190	130
Strong demand, high cost	0.135	200	130	50
Steady demand, low cost	0.275	160	110	60
Steady demand, high cost	0.225	100	50	-20
Weak demand, low cost	0.110	50	40	10
Weak demand, high cost	0.090	-10	-20	-70

Applying the Maximax criterion We select the largest possible profit for each strategy.

$$\text{máx}_i \, \text{Payoff}(A, S_i) = 260$$

$$\text{máx}_i \, \text{Payoff}(B, S_i) = 190$$

$$\text{máx}_i \, \text{Payoff}(C, S_i) = 130$$

$$\text{máx}\{260, 190, 130\} = 260$$

The Maximax rule recommends Strategy A.

Applying the Maximin criterion We compare the worst outcomes across strategies.

$$\min_i \text{Payoff}(A, S_i) = -10$$
$$\min_i \text{Payoff}(B, S_i) = -20$$
$$\min_i \text{Payoff}(C, S_i) = -70$$

$$\max\{-10, -20, -70\} = -10$$

The Maximin rule recommends Strategy A.

Formulating the decision problem for an optimal choice Expected profits are computed with the combined-state probabilities.

$$\begin{aligned}
EV_A &= 0{,}165(260) + 0{,}135(200) + 0{,}275(160) + 0{,}225(100) + 0{,}110(50) + 0{,}090(-10) \\
&= 42{,}9 + 27 + 44 + 22{,}5 + 5{,}5 - 0{,}9 = 141{,}0 \\
EV_B &= 0{,}165(190) + 0{,}135(130) + 0{,}275(110) + 0{,}225(50) + 0{,}110(40) + 0{,}090(-20) \\
&= 31{,}35 + 17{,}55 + 30{,}25 + 11{,}25 + 4{,}4 - 1{,}8 = 93{,}0 \\
EV_C &= 0{,}165(130) + 0{,}135(50) + 0{,}275(60) + 0{,}225(-20) + 0{,}110(10) + 0{,}090(-70) \\
&= 21{,}45 + 6{,}75 + 16{,}5 - 4{,}5 + 1{,}1 - 6{,}3 = 35{,}0
\end{aligned}$$

$$\max\{141{,}0, 93{,}0, 35{,}0\} = 141{,}0$$

The expected value criterion recommends Strategy A.

Problem 3: Structured solution

Interpreting decision alternatives, events, consequences, and states of nature

Element	Description
Decision alternatives	Model A (standard), Model B (premium), Model C (flex)
States of nature	Combined demand and staffing-cost outcomes (6 states)
Events	Demand and staffing-cost conditions observed after selection
Consequences	Profit (dollars) for each model–state pair
Probabilities	Demand: 0.25, 0.50, 0.25; Staffing: 0.60, 0.40; independent

Building the payoff table from the narrative data Using profit = (members × fee) − (members × variable cost) − fixed cost.

State of nature	Probability	Model A	Model B	Model C
High demand, favorable cost	0.15	4040	5400	3200
High demand, unfavorable cost	0.10	-920	1560	-2400
Medium demand, favorable cost	0.30	1320	1800	1040
Medium demand, unfavorable cost	0.20	-2360	-1080	-3120
Low demand, favorable cost	0.15	-1400	-1800	-880
Low demand, unfavorable cost	0.10	-3800	-3720	-3760

Applying the Maximax criterion We compare the best-case profits for each model.

$$\max_i \text{Payoff}(A, S_i) = 4040$$
$$\max_i \text{Payoff}(B, S_i) = 5400$$
$$\max_i \text{Payoff}(C, S_i) = 3200$$

$$\max\{4040, 5400, 3200\} = 5400$$

The Maximax rule recommends Model B.

Applying the Maximin criterion We compare the worst outcomes for each model.

$$\min_i \text{Payoff}(A, S_i) = -3800$$
$$\min_i \text{Payoff}(B, S_i) = -3720$$
$$\min_i \text{Payoff}(C, S_i) = -3760$$

$$\max\{-3800, -3720, -3760\} = -3720$$

The Maximin rule recommends Model B.

Formulating the decision problem for an optimal choice Expected profits use the combined-state probabilities.

$$\begin{aligned} EV_A &= 0{,}15(4040) + 0{,}10(-920) + 0{,}30(1320) + 0{,}20(-2360) + 0{,}15(-1400) + 0{,}10(-3800) \\ &= 606 - 92 + 396 - 472 - 210 - 380 = -152 \\ EV_B &= 0{,}15(5400) + 0{,}10(1560) + 0{,}30(1800) + 0{,}20(-1080) + 0{,}15(-1800) + 0{,}10(-3720) \\ &= 810 + 156 + 540 - 216 - 270 - 372 = 648 \\ EV_C &= 0{,}15(3200) + 0{,}10(-2400) + 0{,}30(1040) + 0{,}20(-3120) + 0{,}15(-880) + 0{,}10(-3760) \\ &= 480 - 240 + 312 - 624 - 132 - 376 = -580 \end{aligned}$$

$$\max\{-152, 648, -580\} = 648$$

The expected value criterion recommends Model B.